Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/01/2015
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Altered Ocular Structure/Function and Mitigation of Peri-Optic Nerve Edema During Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
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Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 NSBRI-RFA-12-02 Postdoctoral Fellowships
Start Date:	12/01/2012	**End Date:**	05/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2015 per NSBRI (Ed., 9/16/15) NOTE: End date changed to 11/30/2015 per NSBRI submission (Ed., 12/12/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hargens, Alan Ph.D. (MENTOR/ University of California at San Diego)		
Grant/Contract No.:	NCC 9-58-PF03001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

POSTDOCTORAL FELLOWSHIP
This project directly addresses Critical Path Roadmap Risks and Questions regarding "Risk of Microgravity-Induced Visual Alterations and Intracranial Pressure", specifically IRP Gap CV7: "How are fluids redistributed in-flight?" and IRP Gap VIIP1: "What is the etiology of visual acuity and ocular structural and functional changes seen in-flight and post-flight?" and IRP Gap VIIP3: "What in-flight diagnostic tools are needed to measure changes in intraocular pressure and intracranial pressure?" Dr. Macias' primary work over this period has focused on understanding intraocular pressure (IOP) and noninvasive intracranial pressure (ICP) in ground-based tilt studies. The focus of this project is to determine if lower body negative pressure (LBNP) used during simulated spaceflight impacts ICP, ocular structure and function, and IOP using state-of-the-art techniques. Additionally, because some methods, hardware, and protocols employed as part of this project are the same as our Fluid Shifts experiment with Chibis-LBNP, we can make direct comparisons to the International Space Station (ISS) Fluid Shifts experiment.

SUPPLEMENTAL INFORMATION FOR FINAL REPORT--NOVEMBER 2015: Dr. Macias' early work focused on determining a whole body tilt angle to generate a significant changes in IOP and ICP. In addition, it was important to determine the minimum length of time required to generate significant changes in IOP and ICP during tilt and LBNP exposure. Twenty-five normal healthy non-smoking volunteers participated in this study. Subjects were positioned supine (5 mins), sitting (5 mins), 15 whole body head down tilt (HDT) (5 mins) and ten minutes of HDT with LBNP (25 mmHg). The order of HDT and HDT+LBNP tests were balanced. Right and left IOP, transcranial ultrasound pulse amplitude (noninvasive ICP surrogate), arm blood pressure, and heart rate were measured during the last minute (steady state) of each testing condition. IOP significantly decreased from supine to sitting posture by 3.2 ± 1.4 mmHg (mean ± standard deviation), and increased by 0.9 ± 1.3 mmHg from supine to the HDT position. LBNP during HDT significantly lowered IOP to supine levels. In addition, LBNP significantly reduced transcranial ultrasound pulse amplitudes (noninvasive ICP) by 38% as compared to the HDT condition (n=9). These data are now published in the journal Aerospace Medicine and Human Performance.

During the third year of Dr. Macias' fellowship he has enrolled 15 subjects into the Cerebral and Cochlear Fluid Pressure (CCFP, a noninvasive ICP surrogate) study. Note, a decrease in tympanic membrane displacement as measured by the CCFP corresponds to an increase in ICP. Noninvasive CCFP measures of ICP significantly increased from the sitting to the supine posture ($p<0.001$). Moreover, ICP further increased significantly when the subjects were positioned in the HDT position, as compared to supine ($p<0.001$). However, when LBNP (25 mmHg and 50 mmHg) was activated during HDT, ICP decreased significantly ($p=0.002$), as compared to HDT. Moreover, cross sectional area measures of the internal jugular vein parallel these noninvasive ICP findings. Therefore, LBNP may reduce cephalad venous congestion during simulated microgravity and reduce ICP. These data provide further evidence that LBNP may be an effective countermeasure to mitigate cephalad fluid shifts.

DECEMBER 2014 REPORT: Dr. Macias' early work focused on determining a whole body tilt angle to generate significant changes in IOP and ICP. In addition, it was important to determine the minimum length of time required to generate significant changes in IOP and ICP during tilt and LBNP exposure. Twenty-five normal healthy non-smoking volunteers participated in this study (mean age: 36 years). Subjects were positioned supine (5 mins), sitting (5 mins), 15 whole body head-down-tilt (HDT) (5 mins), and 10 minutes of HDT with LBNP (25 mmHg). The order of HDT and HDT+LBNP tests were balanced. Right and left IOP, transcranial ultrasound pulse amplitude (noninvasive ICP surrogate), arm blood pressure, and heart rate were measured during the last minute (steady state) of each testing condition. IOP significantly decreased from supine to sitting posture by 3.2 ± 1.4 mmHg (mean ± standard deviation), and increased by 0.9 ± 1.3 mmHg from supine to the HDT position. LBNP during HDT significantly lowered IOP to supine levels. In addition, LBNP significantly reduced transcranial ultrasound pulse amplitudes (noninvasive ICP) by 38% as compared to the HDT condition (n=9). The data collected during his fellowship has resulted in a manuscript in press, and eight abstracts.

In year 2, the research team has enrolled 17 subjects into the Cerebral and Cochlear Fluid Pressure (CCFP, a noninvasive ICP surrogate) study. Of the 17 subjects, 10 demonstrate a response to LBNP during HDT and 7 did not respond to the LBNP countermeasure. In the responder cohort, noninvasive CCFP measures of ICP significantly increased from the sitting to the supine posture. Moreover, ICP further increased significantly when the subjects were positioned in the HDT position. However, when LBNP was activated during HDT, ICP decreased significantly.

In the next 6-months, Dr. Macias plans to call back the non-responding subjects to repeat these experiments to determine if indeed these subjects do not respond to LBNP. The CCFP (evoked measure) is not a continuous measurement and measures can be confounded, for example, by subject movement or poor ear seal. However, these data provide important evidence that LBNP may be an effective countermeasure to mitigate cephalad fluid shifts. Therefore, it will be crucial to repeat these CCFP measurements of the non-responding subjects. In addition, Dr. Macias plans to measure optic nerve sheath diameter and facial soft tissue thickness using a standard ultrasound device. Dr. Macias' future work will determine if fluid shift countermeasures (thigh cuffs and LBNP) reduce invasive measures of ICP during simulated microgravity.

Rationale for HRP Directed Research:

Earth Benefits: These studies defined how simulated microgravity affects fluid redistribution in brain and eye. Understanding how pressures of the brain and eye interact during simulated spaceflight will support human space exploration and enhance life on Earth of common clinical eye and brain pathologies through advances in space biomedical science and technology. Management of intracranial hypertension, intracranial pressure, and pressure related vision pathology requires novel noninvasive management strategies. Lower body negative pressure (LBNP) is a device that shifts fluid away from the head and into lower body regions. Data obtained as part of this project suggest that LBNP reduces internal jugular vein cross sectional area, intraocular pressure, and intracranial pressure. Therefore, LBNP may provide clinicians with a new tool to manage intracranial congestion and reduce swelling of the optic nerve head region. Research Impact: With respect to research productivity during 2012-2015, Dr. Macias published 11 peer-reviewed papers in the areas of orthopaedics, fracture healing, bone biology, and space physiology. In addition, he also presented many abstracts, some of which he or his mentees received major awards at national and international meetings. Some service contributions to University of California San Diego (UCSD) and U.S. public organizations are noteworthy during 2012-2015. During 2012-2015, Dr. Macias mentored and professionally advanced a disadvantaged-minority student in the lab through a summer research grant, 2015-APS Short-Term Research Education Program to Increase Diversity in Health-Related Research (STRIDE) Undergraduate Summer Research Fellowship to Amarachi Uzosike: To

Research Impact/Earth Benefits:	support a summer research opportunity in the Clinical Physiology Research Laboratory titled, Hemodynamics during simulated compartment syndrome. Shannon Baird, a UCSD/Preuss School science teacher (unique charter middle and high school for low income students who strive to become the first in their families to graduate from college) was awarded a 2014 American Physiological Society Research Teacher Professional Development Fellowship to work with Dr. Macias on Fluid Shift related experimental protocols. He has been an active postdoctoral mentor to medical, undergraduate students, and a teacher. In terms of teaching/mentorship, his students in the Department of Orthopaedic Surgery have been instructed and guided such that eight of them have won significant honors during instructional and research experience during 2012-2015. In terms of service, Dr. Macias organized an event in coordination with NASA titled, Destination Station, with the primary aim to educate UCSD faculty on research funding opportunities on the International Space Station. In addition, he served as a panelist for 17-high school students for UCSD-Academic Connections Career Development Program. Other service included an outreach event titled, Spaceflight Physiology Understanding Week, in Shannon Baird's class at The Preuss School. In terms of public service, Dr. Macias served as an Undergraduate Research Judge for the Experimental Biology David Bruce Award. In addition, to the mentorship of undergraduate, graduate, and medical students in our lab, Dr. Macias co-taught a UCSD Seminar Space Physiology and Exploration (WARR 87) with Dr. Alan Hargens.
Task Progress:	SUPPLEMENTAL INFORMATION FOR FINAL REPORT--NOVEMBER 2015: Twenty-five normal healthy non-smoking volunteers participated in this study. Subjects were positioned supine (5 mins), sitting (5 mins), 15 whole body HDT (5 mins), and ten minutes of HDT with LBNP (25 mmHg). The order of HDT and HDT+LBNP tests was balanced. Right and left IOP, transcranial ultrasound pulse amplitude (noninvasive ICP surrogate), arm blood pressure, and heart rate were measured during the last minute (steady state) of each testing condition. IOP significantly decreased from supine to sitting posture by 3.2 ± 1.4 mmHg (mean ± standard deviation), and increased by 0.9 ± 1.3 mmHg from supine to the HDT position. LBNP during HDT significantly lowered IOP to supine levels. In addition, LBNP significantly reduced transcranial ultrasound pulse amplitudes (noninvasive ICP) by 38% as compared to the HDT condition (n=9). These data are now published in the journal Aerospace Medicine and Human Performance. During the third year of Dr. Macias' fellowship he has enrolled 15 subjects into the Cerebral and Cochlear Fluid Pressure (CCFP, a noninvasive ICP surrogate) study. Note, a decrease in tympanic membrane displacement as measured by the CCFP corresponds to an increase in ICP. Noninvasive CCFP measures of ICP significantly increased from the sitting to the supine posture ($p<0.001$). Moreover, ICP further increased significantly when the subjects were positioned in the HDT position, as compared to supine ($p<0.001$). However, when LBNP (25 mmHg and 50 mmHg) was activated during HDT, ICP decreased significantly ($p=0.002$), as compared to HDT. Cross sectional area measures of the internal jugular vein parallel these noninvasive ICP findings. Therefore, LBNP may reduce cephalad venous congestion during simulated microgravity and reduce ICP. These data provide further evidence that LBNP may be an effective countermeasure to mitigate cephalad fluid shifts. DECEMBER 2014 REPORT: Recent ophthalmic evaluations of astronauts after their 6-month missions to the International Space Station (ISS) revealed unexpected vision problems (NASA, 2011; Mader et al., 2011). After their long-duration spaceflights, 7 astronauts presented anomalies including decreased near vision in 6, globe flattening in 5, optic disc edema in 5, choroidal folds in 5, retinal nerve fiber layer thickening in 6, and cotton wool spots within the retinal nerve fiber layer in 3. A more recent study by NASA retrospectively reviewed crew medical records to determine the incidence of vision impairment. Lumbar punctures performed in the 4 astronauts with optic disc edema showed moderately elevated CSF pressure (16-21 mmHg, measures taken 12-66 days post-landing). While there are many possible explanations for these vision problems in astronauts, elevated ICP due to a headward fluid shift during microgravity (NASA, 2011; Watenpaugh and Hargens, 1996) is proposed as the leading mechanism for the observed disc edema, globe flattening, choroidal folds, and hyperopic shifts (Mader et al., 2011; Marshall-Bowman et al., 2013). We hypothesize that a fluid shift countermeasure will reduce intracranial pressure (ICP) during simulated microgravity exposure to supine levels. We will test proven headward fluid shift countermeasures (e.g., lower body negative pressure (LBNP) and Braslet) to mitigate elevations of ICP and IOP. The same device used for pre-, in-, and post-flight ICP measures of astronauts will be utilized. These studies will employ a whole body tilt protocol known to induce headward fluid shifts and raise ICP, enabling direct comparisons of IOP, ICP, and ocular structures-function. Moreover, these studies will determine the efficacy of lower body negative pressure or Braslet countermeasures to mitigate elevations in ICP and IOP during conditions known to induce headward fluid shifts. During year-2 we completed the intraocular pressure and ICP (utilizing the Pulsed Phase Lock Loop device) studies during whole body tilt with LBNP. These data have resulted in a article in press (Macias BR, Grande-Gutierrez N, JHK Liu, and AR Hargens. Intraocular and Intracranial Pressures during Head-Down-Tilt with Lower Body Negative Pressure. Aviation Space and Environmental Medicine, In Press). This manuscript will be published under the new journal title "Aerospace Medicine and Human Performance." It is scheduled for the inaugural issue in January 2015. In addition, we have enrolled seventeen subjects into the Cochlear Fluid Pressure (CCFP, a noninvasive ICP surrogate) study. Currently, we are analyzing the CCFP data generated from this study. We have successfully renewed our IRB approval and amended the protocol to include ultrasound measures (vascular and optic nerve sheath diameter) and facial soft tissue thickness during tilt and LBNP.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
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